

Lec 1-2 (Sept 3, 2025)

- 1. Limits (2.3)
- 2.
 - Continuity (2.5)
 - Limit properties / Limit Laws (2.2)
 - Composition of continuous functions (2.5)
 - Limit computation (basic)

- 3.
 - One-sided limits (2.4)
 - One-sided continuity (2.4)
- 4.
 - Discontinuities (2.5)
 - Continuous extension (2.5)
- 5. Intermediate value theorem (2.5)

- 6. Limits involving bounded functions (2.2, extended)
- 7. One-sided continuity and limits (2.4, extended)

In today's lecture, red block means "optional and skipped at this moment" and black block means "definitions, theorems and other important issues".

There are a lot of contents. Try your best to digest: confident, humble, peer

Comments. Examples are important. Math theory are originated from and generalized from concrete examples; instead of coming out from nowhere.

§1.4. Formal Definition. [optional]

Def. Let $f: D \rightarrow \mathbb{R}$ be a function defined on an open interval containing c , except possibly at c itself (i.e., c may be not in the domain like case ③).

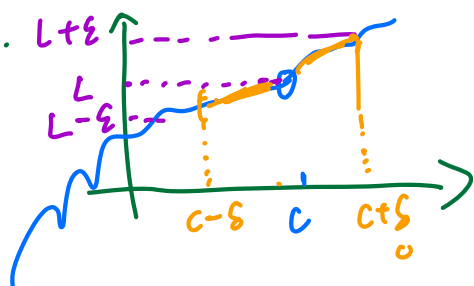
Let $L \in \mathbb{R}$. Then we say "the limit of f at c exists and equals L " and write $\lim_{x \rightarrow c} f(x) = L$ if for any $\varepsilon > 0$, there exists a $\delta > 0$ such that for all $x \in D$ with $0 < |x - c| < \delta$, we have $|f(x) - L| < \varepsilon$.

$$x \in (c - \delta, c) \cup (c, c + \delta)$$

$f(x) \in (L - \varepsilon, L + \varepsilon)$: $f(x)$ is close to L in a neighborhood.

"in the neighborhood, all values coincide the trend".

Illustration.



① Let ε be any arbitrarily small value (e.g., $\varepsilon = 0.01$)

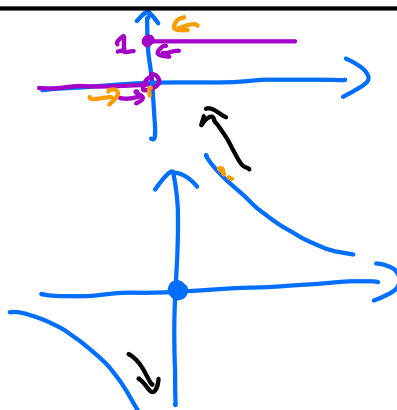
② Find a $\delta > 0$, s.t. $|f(x) - L| < \varepsilon$ on $x \in (c - \delta, c + \delta) \setminus \{c\} \cap D$.

• Go back and examine the above three graphs.

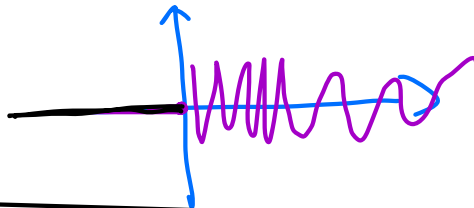
§1.5. counter examples: $\lim_{x \rightarrow 0} f(x)$ does not exist.

Case 1. $f(x) = \begin{cases} 0, & x < 0 \\ 1, & x \geq 0 \end{cases}$

Case 2. $f(x) = \begin{cases} \frac{1}{x}, & x \neq 0 \\ 0, & x = 0 \end{cases}$



Case 3. $f(x) = \begin{cases} 0, & x \leq 0 \\ \sin \frac{1}{x}, & x > 0 \end{cases}$



Reasons:

Case 1: approaching different numbers from different sides:

$$f(x) \rightarrow 0 \text{ as } x \rightarrow 0 \text{ from left.}$$

$$f(x) \rightarrow 1 \text{ as } x \rightarrow 0 \text{ from right.}$$

[Formally, suppose $\lim_{x \rightarrow 0} f(x)$ exists and equals L . Then L should be 0 or 1.

Take $L=0$ as an example. We set $\epsilon=0.5$.

Assume by contradiction that ^(there exists) $\exists \delta > 0$ st. $|f(x)-0| < 0.5$ for any $x \in (0-\delta, 0+\delta) \setminus \{0\}$.

But it is not true because if $x \in (0, 0+\delta)$, $|f(x)-0| = |1-0| = 1 > 0.5$.

It is an contradiction!

Hence, $\lim_{x \rightarrow 0} f(x)$ doesnot exist.]

Case 2: As x approaches 0, $f(x)$ can be arbitrarily large or small

[so L is not finite].

Case 3. Oscillating from the right between two numbers, $(-1, 1)$

i.e. At any neighborhood of 0, there exists two points x_0 and x_1 ,

$$\text{s.t. } \begin{cases} |x_0-0| < \delta \text{ and } |x_1-0| < \delta \\ \sin(\frac{1}{x_0}) = 1 \text{ and } \sin(\frac{1}{x_1}) = -1 \end{cases} \quad \begin{cases} \frac{1}{x_0} = \frac{\pi}{2} + 2k\pi \Rightarrow x_0 = \frac{1}{\frac{\pi}{2} + 2k\pi}, k=1, 2, 3, \dots \\ \frac{1}{x_1} = \frac{3\pi}{2} + 2k\pi \Rightarrow x_1 = \frac{1}{\frac{3\pi}{2} + 2k\pi} \end{cases}$$

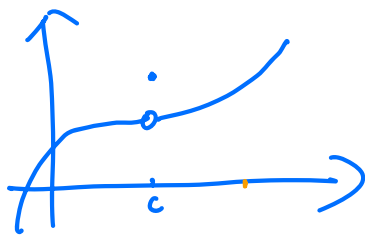
$$\text{making } |\sin(\frac{1}{x_0}) - L| = |1-L| < \epsilon \text{ and } |\sin(\frac{1}{x_1}) - L| = |-1-L| < \epsilon$$

not hold at the same time.

Alternative notation. We can write $f(x) \rightarrow L$ as $x \rightarrow c$ to represent $\lim_{x \rightarrow c} f(x) = L$.

§2. Continuity

"Discontinuous point":



In general,

$$\lim_{x \rightarrow c} f(x) \neq f(c).$$

§2.1. Formal definition

Def: Let $f: D \rightarrow \mathbb{R}$ a function defined on an open interval containing c . We say that f is continuous at c if

$$\lim_{x \rightarrow c} f(x) = f(c)$$

"the trend is the value".

A function defined on an open interval D is said to be continuous if it is continuous at each point in D .

a point



e.g. The following functions we are familiar with are continuous (on domain D):

- Constant function: $f(x) = k$, where k is constant, $D = \mathbb{R}$.

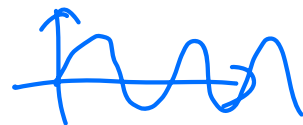
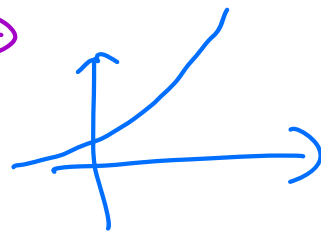
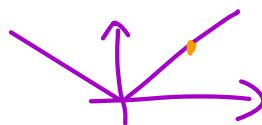
- Identity function: $f(x) = x$, $D = \mathbb{R}$.

- Absolute value function: $f(x) = |x|$, $D = \mathbb{R}$.

- Natural exponential function: $f(x) = e^x$, $D = \mathbb{R}$.

- Natural logarithmic function: $f(x) = \ln x$, $D = (0, \infty)$.

- Basic trigonometric function: $f(x) = \sin x$, $g(x) = \cos x$, $D = \mathbb{R}$ for both.



Comment: The names of math terminology coincide with the natural shapes or meanings.
For example, continuity, limit, neighborhood.

• By definition, limit of a continuous function can be computed by direct substitution

for any point in the domain: $\lim_{x \rightarrow 1} \sin x = \sin 1$.

§2.2. Limit Laws

The following limit laws may make computations more convenient.

Thm 2.2.1 [Limit Laws] If L, M, c and k are real numbers and

$\lim_{x \rightarrow c} f(x) = L$ and $\lim_{x \rightarrow c} g(x) = M$, then

1. Sum rule: $\lim_{x \rightarrow c} (f(x) + g(x)) = \lim_{x \rightarrow c} f(x) + \lim_{x \rightarrow c} g(x) = L + M$.

2. Difference rule: $\lim_{x \rightarrow c} (f(x) - g(x)) = L - M$.

3. Constant multiple rule: $\lim_{x \rightarrow c} (k \cdot f(x)) = k \cdot L$.

4. Product rule: $\lim_{x \rightarrow c} (f(x) \cdot g(x)) = L \cdot M$.

5. Quotient rule: $\lim_{x \rightarrow c} \frac{f(x)}{g(x)} = \frac{\lim_{x \rightarrow c} f(x)}{\lim_{x \rightarrow c} g(x)} = \frac{L}{M}$, $M \neq 0$.

6. Power rule: $\lim_{x \rightarrow c} [f(x)]^n = L^n$, n is a positive integer

7. Root rule: $\lim_{x \rightarrow c} \sqrt[n]{f(x)} = \sqrt[n]{L} = L^{1/n}$, n is a positive integer

(If n is even, we assume that $\lim_{x \rightarrow c} f(x) = L > 0$.)

By Limit laws, limits of algebraic combinations (+, -, $\times$, $\div$, $(\cdot)^n$, $\sqrt{\cdot}$) of continuous function can also be computed by direct substitution (for points in domains)

e.g. $\lim_{x \rightarrow 2} \frac{\sqrt{\ln x}}{x^2 - 3}$ laws $\frac{\lim_{x \rightarrow 2} \sqrt{\ln x}}{\lim_{x \rightarrow 2} (x^2 - 3)}$

laws 7 & 2 $\frac{\sqrt{\lim_{x \rightarrow 2} \ln x}}{\lim_{x \rightarrow 2} x^2 - \lim_{x \rightarrow 2} 3}$

continuity $\frac{\sqrt{\ln 2}}{2^2 - 3} = \sqrt{\ln 2} \cdot \boxed{\frac{1}{4}}$

Def. A polynomial is a function of the form

$$p(x) = a_n x^n + a_{n-1} x^{n-1} + \dots + a_1 x + a_0,$$

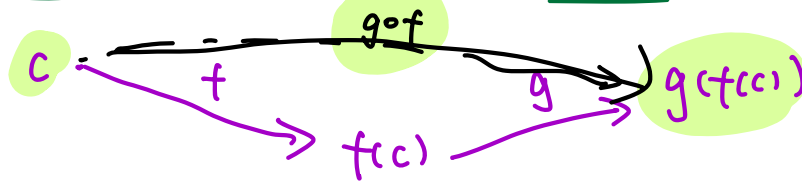
where $a_0, a_1, \dots, a_n$ are constants. A rational function is a function of the form $\frac{p(x)}{q(x)}$

where $p(x)$ and $q(x)$ are both polynomials (and $q(x)$ is not the zero function).

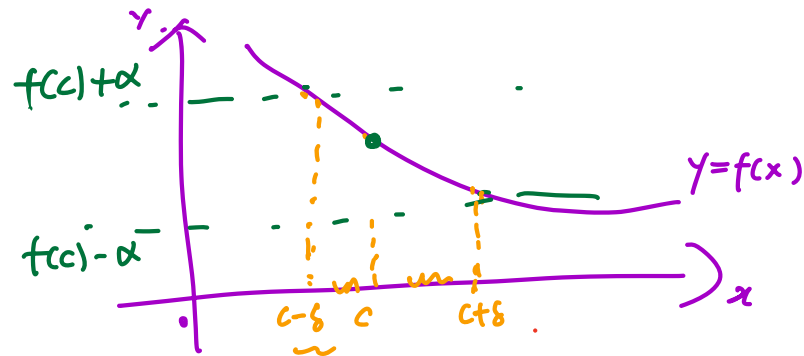
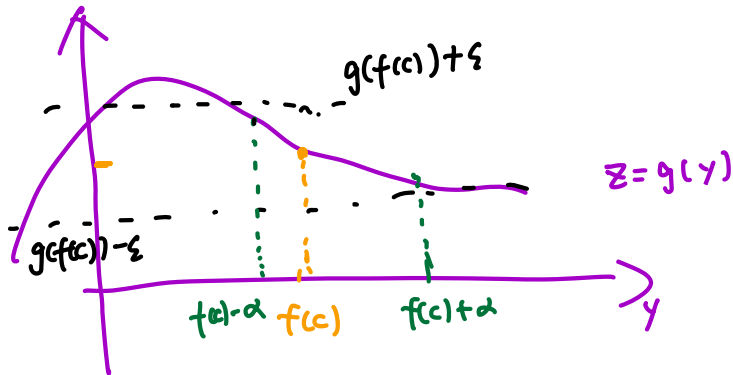
Rmk. Since polynomial and rational functions are finite algebraic combinations of continuous functions $f(x)=k$ and $g(x)=x$, by limit laws, their limits (for points in the domain) can be found by direct substitutions. Hence, they are continuous.

§2.3. Composition and continuity

Thm 2.5.9. If f is continuous at C and g is continuous at $f(C)$, then $g \circ f$ is continuous at C .



Pf: [Optional]



We show that $\lim_{x \rightarrow C} (g \circ f)(x) = (g \circ f)(C)$.

Let $z = g(y)$, $y = f(x)$, so $z = (g \circ f)(x)$.

Let $\epsilon > 0$. Since g is continuous at $f(C)$, $\exists \alpha > 0$ s.t.

$$|g(y) - g(f(C))| < \epsilon \quad \text{for any } y \in \left((f(C) - \alpha, f(C) + \alpha) \setminus \{f(C)\} \right) \cap \text{domain}(g).$$

Since f is continuous at C , for this $\alpha > 0$, $\exists \delta > 0$, s.t.

$$|f(x) - f(c)| < \underline{\alpha} \text{ for any } x \in (c-\delta, c+\delta) \setminus \{c\}$$

Hence, for any $\varepsilon > 0$, $\exists \delta > 0$, s.t. $\cap \text{domain}(f)$.

$$|(g \circ f)(x) - (g \circ f)(c)| < \varepsilon \text{ for any } x \in (c-\delta, c+\delta) \setminus \{c\}$$

Hence, $\lim_{x \rightarrow c} (g \circ f)(x) = (g \circ f)(c)$. $\cap \text{domain}(f)$ k.

• In fact, a more general result holds.

Thm 2.5.10. If g is continuous at b , and $b = \lim_{x \rightarrow c} f(x)$, then

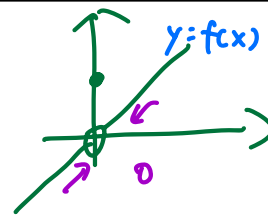
$$\lim_{x \rightarrow c} g(f(x)) = g(b) = g\left(\lim_{x \rightarrow c} f(x)\right).$$

Its proof is similar. Thm 2.5.10 means that you may bring the "lim" inside the $\lim_{x \rightarrow c}$ outside function whenever it is continuous (provided that the inside limit exists).

e.g. $g(x) = \sin x$, $f(x) = \begin{cases} 1, & \text{if } x=0, \\ x, & \text{if } x \neq 0. \end{cases}$

[outside function, continuous]

[inside function, not continuous]



By thm 2.5.10, since g is continuous,

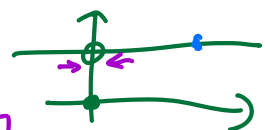
$$\lim_{x \rightarrow 0} g(f(x)) = g\left(\lim_{x \rightarrow 0} f(x)\right) = g(0) = \sin 0 = 0.$$

Note that you cannot do the same in general if the "outside" function is not continuous.

e.g. Let $f(x) = x$ and $g(x) = \begin{cases} 0, & \text{if } x=0 \\ 1, & \text{if } x \neq 0 \end{cases}$

[inside function]

[outside function, not continuous]



$$\therefore \lim_{x \rightarrow 0} g(f(x)) = \lim_{x \rightarrow 0} g(x) = 1. \quad \lim_{x \rightarrow 0} \underline{g(f(x))} \neq g(\lim_{x \rightarrow 0} f(x))$$

But $g(\lim_{x \rightarrow 0} f(x)) = g(0) = 0$. Two quantities are not equal.

§2.4. Limit computation. $\lim_{x \rightarrow 2} \underline{g(f(x))} = g(f(2))$

If we cannot make direct substitution, how to compute the limits?

A) eliminate zero denominator

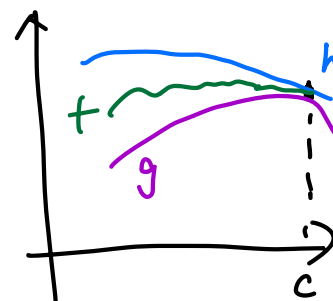
$$\begin{aligned} \text{e.g. } \lim_{x \rightarrow 0} \frac{\sqrt{x^2+100}-10}{x^2} &= \lim_{x \rightarrow 0} \frac{(\sqrt{x^2+100}-10)(\sqrt{x^2+100}+10)}{x^2(\sqrt{x^2+100}+10)} \\ &= \lim_{x \rightarrow 0} \frac{x^2+100-100}{x^2(\sqrt{x^2+100}+10)} \\ &= \lim_{x \rightarrow 0} \frac{x^2}{x^2(\sqrt{x^2+100}+10)} \\ &= \lim_{x \rightarrow 0} \frac{1}{\sqrt{x^2+100}+10} = \frac{1}{20}. \end{aligned}$$

B) Squeeze theorem (i.e., sandwich theorem)

Thm 2.2.4. Suppose that $g(x) \leq f(x) \leq h(x)$ for all x in some open interval containing c , except possibly at $x=c$ itself. Suppose also that

$$\lim_{x \rightarrow c} g(x) = \lim_{x \rightarrow c} h(x) = L.$$

Then $\lim_{x \rightarrow c} f(x) = L$.



e.g. $\lim_{x \rightarrow 0} x^2 \sin \frac{1}{x} = ?$

Step 1: $|x^2 \sin \frac{1}{x}| \leq x^2 \cdot 1 \Rightarrow -x^2 \leq x^2 \sin \frac{1}{x} \leq x^2$

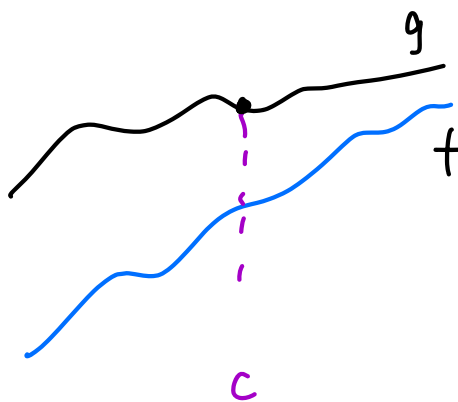
Step 2: $\lim_{x \rightarrow 0} (\pm x^2) = 0 = \lim_{x \rightarrow 0} x^2$

Step 3: By the Sandwich thm, we have $\lim_{x \rightarrow 0} x^2 \sin \frac{1}{x} = 0$. $\square$

Additionally, we have

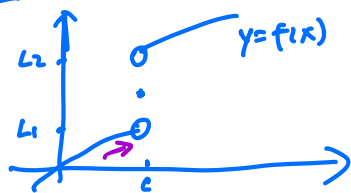
Thm 2.2.5. If $f(x) \leq g(x)$ for all x in some open interval containing c , except possibly at $x=c$ itself, and the limits of f and g both exist as x approaches c , then

$$\lim_{x \rightarrow c} f(x) \leq \lim_{x \rightarrow c} g(x).$$



§3. One-sided limits and continuity

§3.1 One-sided limits



$$\lim_{x \rightarrow c^-} f(x) = L_1, \quad \lim_{x \rightarrow c^+} f(x) = L_2.$$

In general, if f is defined on some interval $(c, c+a)$ for some $a > 0$, then we can talk about right-hand limit:

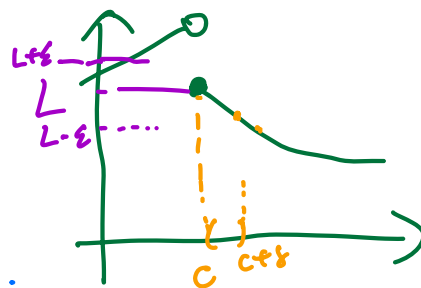
$\lim_{x \rightarrow c^+} f(x)$: "limit of $f(x)$ as x approaches c from the right".

Def: Suppose $f: D \rightarrow \mathbb{R}$ is defined on the interval $(c, c+a)$ for some $a > 0$.

Suppose $L \in \mathbb{R}$. Then we write $\lim_{x \rightarrow c^+} f(x) = L$

if for all $\varepsilon > 0$, there exists a $\delta > 0$, such that

for all $x \in (c, c+\delta)$, we have $|f(x) - L| < \varepsilon$.



Similarly, if f is defined on some interval $(c-a, c)$ for some $a > 0$, then we can talk about the left-hand limit $\lim_{x \rightarrow c^-} f(x)$ and its formal definition is given in Chapter 2.4.

Using the definition of one-sided limits, one can prove that all properties in Thm 2.2.1 still holds for one-sided limits.

$$\text{e.g. } \lim_{x \rightarrow c^+} [f(x) \pm g(x)] = \lim_{x \rightarrow c^+} f(x) \pm \lim_{x \rightarrow c^+} g(x).$$

given that both the one-sided limits on the right hand side exist as finite numbers.

• One can also prove the following theorem using definitions.

Thm 2.4.6. Suppose $f: D \rightarrow \mathbb{R}$ is defined on $(c-a, c+a) \setminus \{c\}$ for some $a > 0$ and

let $L \in \mathbb{R}$. Then

$$\lim_{x \rightarrow c} f(x) = L \iff \lim_{x \rightarrow c^-} f(x) = L = \lim_{x \rightarrow c^+} f(x).$$

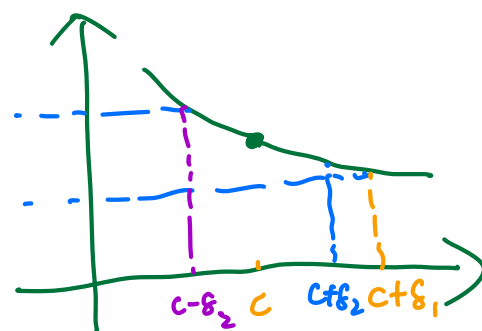
$\forall \varepsilon, \exists \delta, \dots$

for any $x \in (c-\delta, c) \cup (c, c+\delta)$.

Proof [Optional] :

[For interested readers]

$\Rightarrow$ By definition.



$\Leftarrow$ Let $\varepsilon > 0$. There exists $\delta_1 > 0$ and $\delta_2 > 0$ s.t.

$|f(x) - L| < \varepsilon$ for all $x \in D$ and $0 < x - c < \delta_1$ (right-hand limit)

and

$|f(x) - L| < \varepsilon$ for all $x \in D$ and $0 < c - x < \delta_2$. (left-hand limit)

$x \in (c - \delta_2, c)$

Let $\delta \triangleq \min\{\delta_1, \delta_2\}$. We have

$|f(x) - L| < \varepsilon$ for all $x \in D$ and $0 < |x - c| < \delta$,

which means

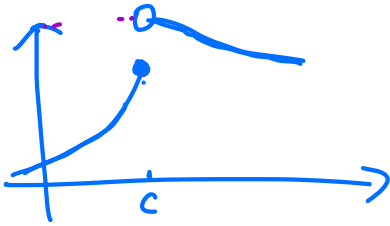
$$\lim_{x \rightarrow c} f(x) = L.$$

□

§ 3.2. One-sided continuity

Definition: • f is left-continuous at c if $\lim_{x \rightarrow c^-} f(x) = f(c)$;
• f is right-continuous at c if $\lim_{x \rightarrow c^+} f(x) = f(c)$.

e.g.



[left trend = function value]

[right trend $\neq$ function value]

§ 4. Discontinuity and continuous extension

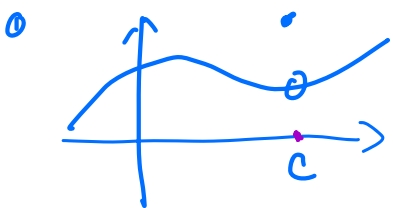
§ 4.1 Discontinuity.

Def. Let f be a function defined on an open interval containing c .

- If f is not continuous at c , then c is called a discontinuity of f
= discontinuous point

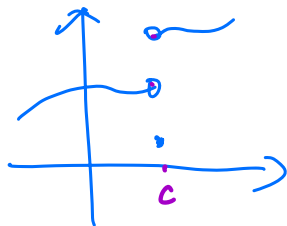
e.g.

removable



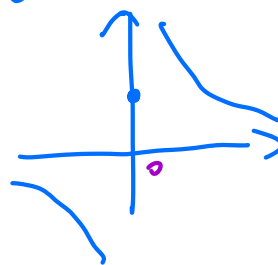
②

jump



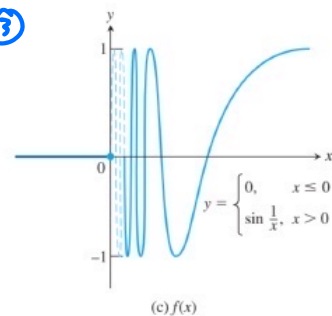
essential (at 0)

③



essential (at 0)

④



① A discontinuity C is said to be **removable** if $\lim_{x \rightarrow C} f(x) = L$ for some $L \in \mathbb{R}$, but $L \neq f(C)$. In this case, f can be made continuous at C if we redefine $f(C) = L$.

② A discontinuity C is called a **jump discontinuity** if

$$\left\{ \begin{array}{l} \lim_{x \rightarrow C^-} f(x) = L_1 \in \mathbb{R}, \\ \lim_{x \rightarrow C^+} f(x) = L_2 \in \mathbb{R}, \\ L_1 \neq L_2 \end{array} \right.$$

③ A discontinuity C is called an **essential discontinuity** if $\lim_{x \rightarrow C^-} f(x)$ or $\lim_{x \rightarrow C^+} f(x)$ does not exist as a finite number

§4.2. Continuous extension

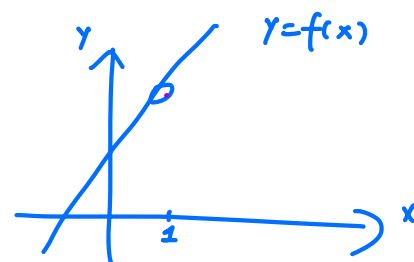
Definition. Let $f: D \rightarrow \mathbb{R}$ be defined near C (but $C \notin D$). If $\lim_{x \rightarrow C} f(x) = L$, where $L \in \mathbb{R}$, then the new function F defined by

$$F(x) \triangleq \begin{cases} f(x), & \text{if } x \in D; \\ L, & \text{if } x = C. \end{cases}$$

is called the **continuous extension** of f to $x = C$.

e.g. For $f(x) = \frac{x^2 - 1}{x - 1}$, we have

$$f(x) = x + 1, \forall x \in \mathbb{R} \setminus \{1\}.$$

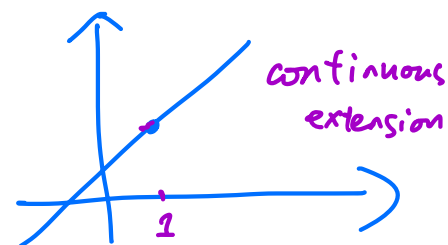


Since , there is a continuous extension

$F: \mathbb{R} \setminus \{1\} \rightarrow \mathbb{R}$ of f to $x=1$:

$$F(x) = \begin{cases} \frac{x^2-1}{x-1} = x+1 & , \text{ if } x \in \mathbb{R} \setminus \{1\} \\ & , \text{ if } x=1 \end{cases}$$

$$= x+1, \forall x \in \mathbb{R}$$

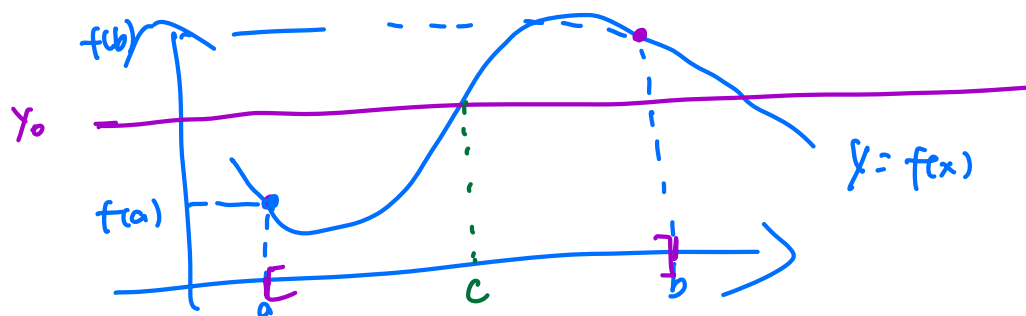


[This is why this discontinuity is called removable.]

§5. Intermediate Value Theorem [IVT]

Thm 2.5.11. [The IVT for continuous functions] [proof requires deeper content]

If f is a continuous function on a closed interval $[a, b]$, and if y_0 is any value between $f(a)$ and $f(b)$, then there exists some $c \in [a, b]$ s.t. $f(c) = y_0$.



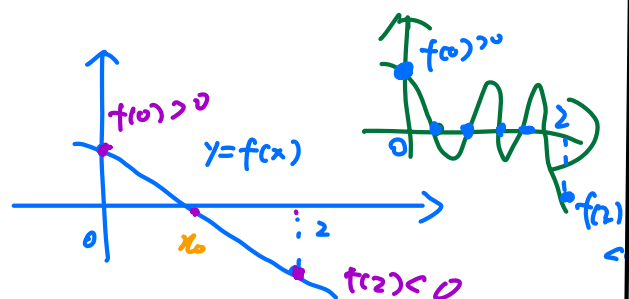
e.g. Show that the equation $\sqrt{2x+5} = 4-x^2$ has a solution.

Pf: Let $f(x) = 4-x^2 - \sqrt{2x+5}$ (continuous!)

• By computation,

$$f(0) = 4 - \sqrt{5} > 0,$$

$$f(2) = -3 < 0.$$



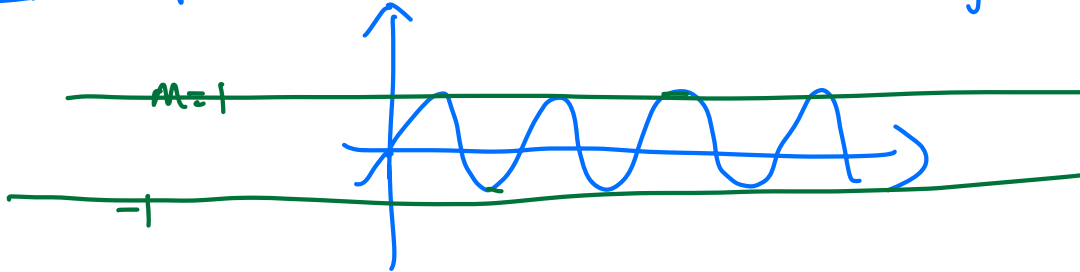
• As $y_0 = 0 \in [-3, 4-\sqrt{5}]$, by IVT, $\exists x_0 \in [0, 2]$ s.t. $f(x_0) = 0$.

$\therefore x = x_0$ is a solution of the above equation.

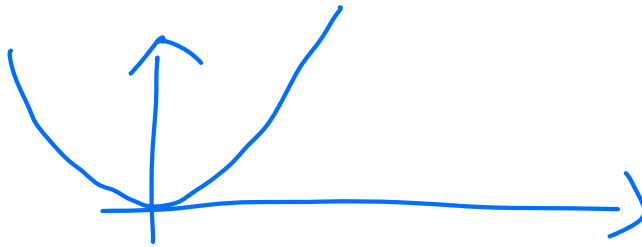
§6. Limits involving bounded functions

Def: A function f is said to be **bounded** on S if there exists $M \in (0, \infty)$ such that $|f(x)| \leq M$ for any $x \in S$.
i.e., $-M \leq f(x) \leq M$

e.g.1. The function $f(x) = \sin x$ is **bounded** on $\mathbb{R}$ (e.g. $M=1$)



e.g.2 The function $f(x) = x^2$ is **not bounded** on $\mathbb{R}$.



Thm (Limits involving bounded functions)

Suppose that f and g are functions defined on some open interval D containing c , except possibly at c [i.e. f and g may be not defined at c].

If $\lim_{x \rightarrow c} f(x) = 0$ and $g(x)$ is bounded on $D \setminus \{c\}$, then

$$\lim_{x \rightarrow c} f(x) \cdot g(x) = 0$$

e.g. $\lim_{x \rightarrow 5} (x^3 - 25x) \cdot \cos(\ln|x-5|) = ?$

Pf: • $\lim_{x \rightarrow 5} (x^3 - 25x) = 0$,

• $|\cos(\ln|x-5|)| \leq 1, \forall x \in \mathbb{R} \setminus \{5\}$.

$\therefore$ By Thm, we have LHS = 0.
(Left-hand side)

□.

Pt of Thm. As g is bounded on $D \setminus \{c\}$,

there exists $M > 0$ such that $|g(x)| \leq M$ for any $x \in D \setminus \{c\}$.

Hence, $|f(x)g(x)| \leq |f(x)| \cdot M, \quad \forall x \in D \setminus \{c\}.$

$\therefore -M|f(x)| \leq f(x)g(x) \leq M \cdot |f(x)|, \quad \forall x \in D \setminus \{c\}. \quad \textcircled{1}$

Because $\lim_{x \rightarrow c} f(x) = 0$, we have

$h(y) \triangleq |y|$



$\lim_{x \rightarrow c} |f(x)| = \lim_{x \rightarrow c} h \circ f(x) = h \left(\lim_{x \rightarrow c} f(x) \right) = \left| \lim_{x \rightarrow c} f(x) \right| = 0.$

↑ here we use the fact that $h(x) = |x|$ is continuous.

$\therefore \lim_{x \rightarrow c} M |f(x)| = M \cdot \lim_{x \rightarrow c} |f(x)| = 0 \quad \textcircled{2}$

similarly, $\lim_{x \rightarrow c} -M |f(x)| = 0 \quad \textcircled{3}$

By $\textcircled{1}, \textcircled{2}, \textcircled{3}$ and the sandwich thm,

we have $\lim_{x \rightarrow c} f(x)g(x) = 0. \quad \textcircled{4}$

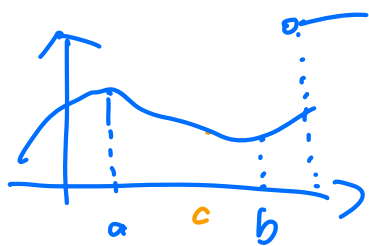
e.g. $f(x) = \begin{cases} x \sin(\frac{1}{x}), & x \neq 0; \\ 0, & x = 0. \end{cases}$

By Thm, $\lim_{x \rightarrow 0} f(x) = \lim_{x \rightarrow 0} x \sin(\frac{1}{x}) = 0 = f(0) \Rightarrow f$ is cont at 0.

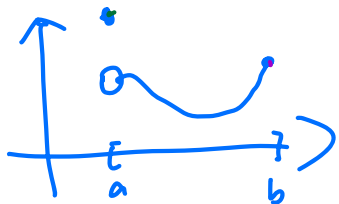
§7. One-sided continuity and limits (continued)

Rmk. A function f is said to be continuous on $[a, b]$ if f is continuous at every $c \in (a, b)$,

left-continuous at b and right-continuous at a .



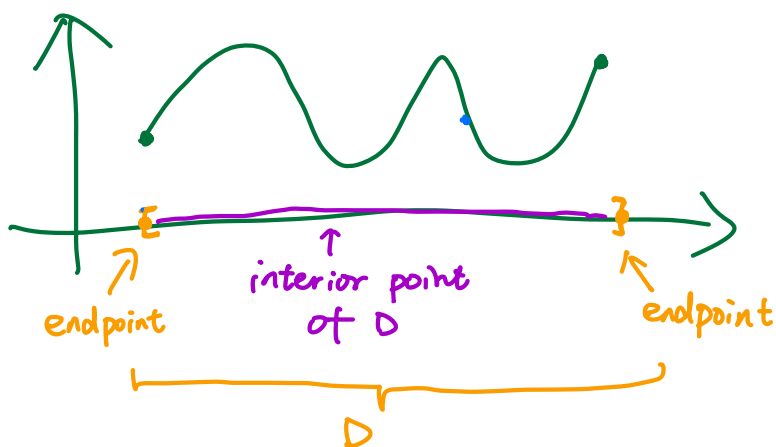
continuous on $[a, b]$



not continuous on $[a, b]$,

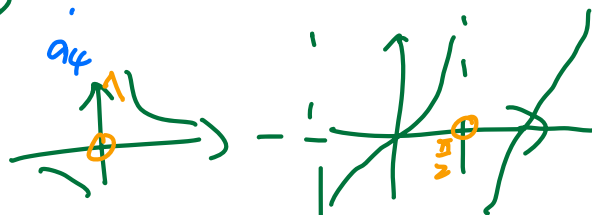
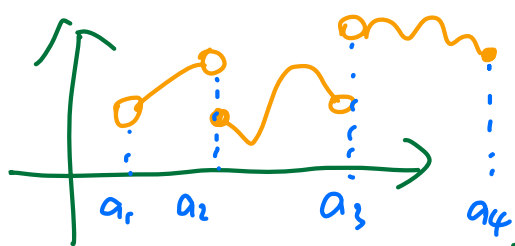
but continuous on $(a, b]$.

- More generally, let $f: D \rightarrow \mathbb{R}$ be a function, where D is a union of intervals. Then f is called a **continuous function** if f is **continuous** at each **interior** point of D and is **one-sided continuous** at each endpoint of D .



(A point $c \in D$ is called an **interior point** of D if there exists $\underline{a} > 0$ such that $(c-a, c+a) \subseteq D$).

e.g. This function is continuous on $(a_1, a_2) \cup (a_2, a_3) \cup (a_3, a_4]$. It is not continuous at a_2 .

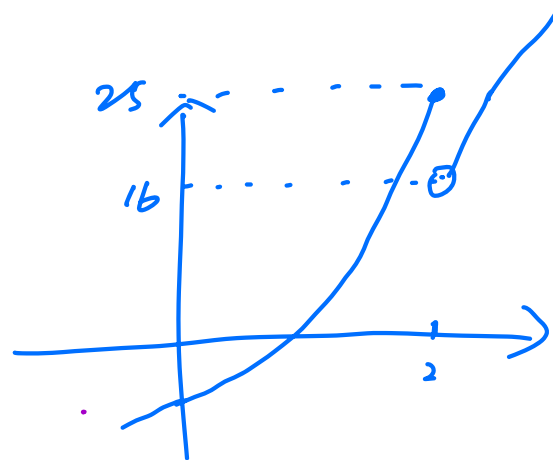


$y = \tan x = f(x)$
 $f(x)$ is cont
 $(-\frac{\pi}{2}, \frac{\pi}{2}) \cup (\frac{\pi}{2}, \frac{3\pi}{2})$

e.g.1. Consider the function $f: \mathbb{R} \rightarrow \mathbb{R}$ defined by

$$f(x) = \begin{cases} 4x+8, & \text{if } x > 2 \\ x^5 - 7, & \text{if } x \leq 2 \end{cases}$$

Then $\lim_{x \rightarrow 2^-} f(x) = 2^5 - 7 = 25$ and $\lim_{x \rightarrow 2^+} f(x) = 4 \cdot 2 + 8 = 16$



by direct substitution (which is valid since f is a piecewise polynomial).

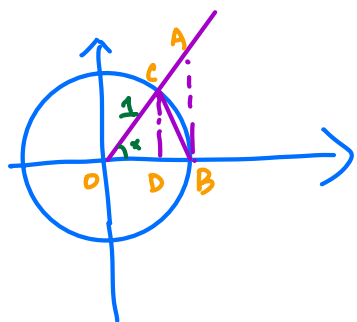
Here, $\lim_{x \rightarrow 2} f(x)$ does not exist. □

e.g.2. A special limit: $\lim_{x \rightarrow 0} \frac{\sin x}{x} = ?$

Pf: We will show that $\lim_{x \rightarrow 0} \frac{\sin x}{x} = \lim_{x \rightarrow 0^+} \frac{\sin x}{x} = 1$.

First, we prove that $\sin x \leq x \leq \tan x$ for $0 < x < \frac{\pi}{2}$.

This is achieved by the circle graph.



Suppose $|\overline{OC}| = |\overline{OB}| = 1$. Let $0 < x < \frac{\pi}{2}$. Then

$$S_{\triangle OCB} = \frac{1}{2} \cdot |\overline{CD}| \cdot |\overline{OB}| = \frac{1}{2} \cdot \sin x \cdot 1.$$

$$S_{\triangle OCB} = \frac{x}{2\pi} \cdot \pi \cdot 1^2 = \frac{1}{2} x.$$

$$S_{\triangle OAB} = \frac{1}{2} \cdot |\overline{AB}| \cdot |\overline{OB}| = \frac{1}{2} \cdot \tan x \cdot 1.$$

Because $|S_{\triangle OCB}| \leq |S_{\triangle OCB}| \leq |S_{\triangle OAB}|$,

we have $\sin x \leq x \leq \tan x$, for any $0 < x < \frac{\pi}{2}$.

Next, $\frac{\sin x}{\sin x} \leq \frac{x}{\sin x} \leq \frac{\tan x}{\sin x}$

$$\Rightarrow 1 \leq \frac{x}{\sin x} \leq \frac{1}{\cos x}$$

$$\Rightarrow \cos x \leq \frac{\sin x}{x} \leq 1, \text{ for any } 0 < x < \frac{\pi}{2}.$$

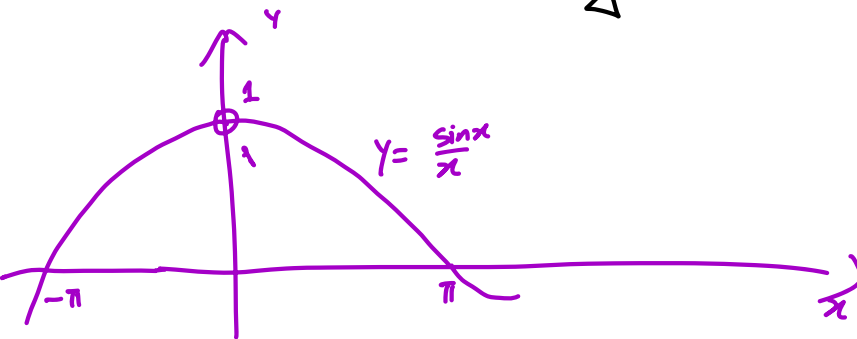
Since $\lim_{x \rightarrow 0^+} 1 = 1 = \lim_{x \rightarrow 0^+} \cos x$, by sandwich theorem,

we have $\lim_{x \rightarrow 0^+} \frac{\sin x}{x} = 1$.

Finally, let $y = -x$. We have $y \rightarrow 0^+$ as $x \rightarrow 0^-$.

$$\begin{aligned} \text{Then } \lim_{x \rightarrow 0^-} \frac{\sin x}{x} &= \lim_{(-y) \rightarrow 0^-} \frac{\sin(-y)}{-y} = \lim_{y \rightarrow 0^+} \frac{-\sin y}{-y} \\ &= \lim_{y \rightarrow 0^+} \frac{\sin y}{y} = 1. \end{aligned}$$

By Thm 2.4.6, $\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$.



e.g. 3. $\lim_{x \rightarrow 0} \frac{\sin 4x}{8x} = \lim_{x \rightarrow 0} \frac{\sin 4x}{4x} \cdot \frac{1}{2} = \lim_{y \rightarrow 0} \frac{\sin y}{y} \cdot \frac{1}{2} = \frac{1}{2}$.

↑
set $y = 4x$. we have $y \rightarrow 0$ as $x \rightarrow 0$.

e.g. 4. $\lim_{x \rightarrow 0} \frac{\cos x - 1}{x} = \lim_{x \rightarrow 0} \frac{\cos(2 \cdot \frac{x}{2}) - 1}{x}$
 $= \lim_{x \rightarrow 0} \frac{1 - 2\sin^2(\frac{x}{2}) - 1}{x}$ [double-angle formula: $\cos(2\theta) = 1 - 2\sin^2\theta$]
 $= \lim_{x \rightarrow 0} \frac{-\sin^2(\frac{x}{2})}{\frac{1}{2}x}$
 $= \lim_{y \rightarrow 0} \frac{-\sin^2 y}{y} \quad [\text{set } y = \frac{x}{2}, y \rightarrow 0 \text{ as } x \rightarrow 0]$

$$= (-1) \lim_{y \rightarrow 0} \frac{\sin y}{y} \lim_{y \rightarrow 0} \sin y \quad [\text{limit law}]$$

$$= -1 \cdot 1 \cdot 0$$

$$= 0$$

eg. 5. $\lim_{t \rightarrow 0} \frac{\tan t \sec(2t)}{3t} \quad [\sec(t) \triangleq \frac{1}{\cos(t)}]$

$$= \lim_{t \rightarrow 0} \frac{\sin t}{3t \cdot \cos t} \cdot \frac{1}{\cos(2t)}$$

$$= \frac{1}{3} \lim_{t \rightarrow 0} \frac{\sin t}{t} \lim_{t \rightarrow 0} \left(\frac{1}{\cos t} \frac{1}{\cos(2t)} \right)$$

$$= \frac{1}{3} \cdot 1 \cdot 1$$

$$= \frac{1}{3}.$$

□